

Task 2: Deploy a Virtual Machine & Install Web Server

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Internship: Cloud Computing Internship

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Task-2 : Deploy a Virtual Machine & Install Web Server

Objective

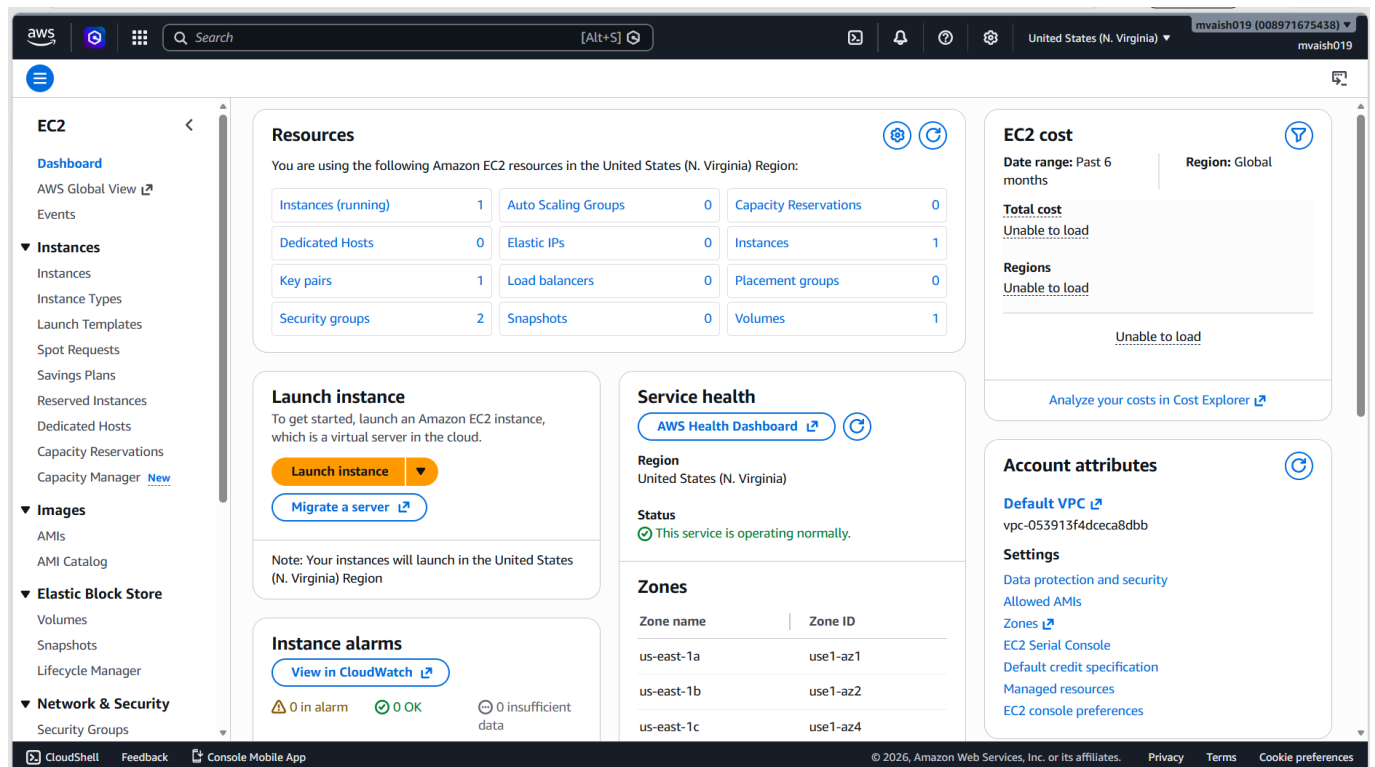
The objective of this task is to create and deploy a Virtual Machine (VM) on AWS EC2, install a web server, and host a simple webpage that can be accessed through a public IP address.

Tools Used

- AWS Management Console
- Amazon EC2
- Amazon Linux 2023
- Apache HTTP Server (httpd)
- Web Browser

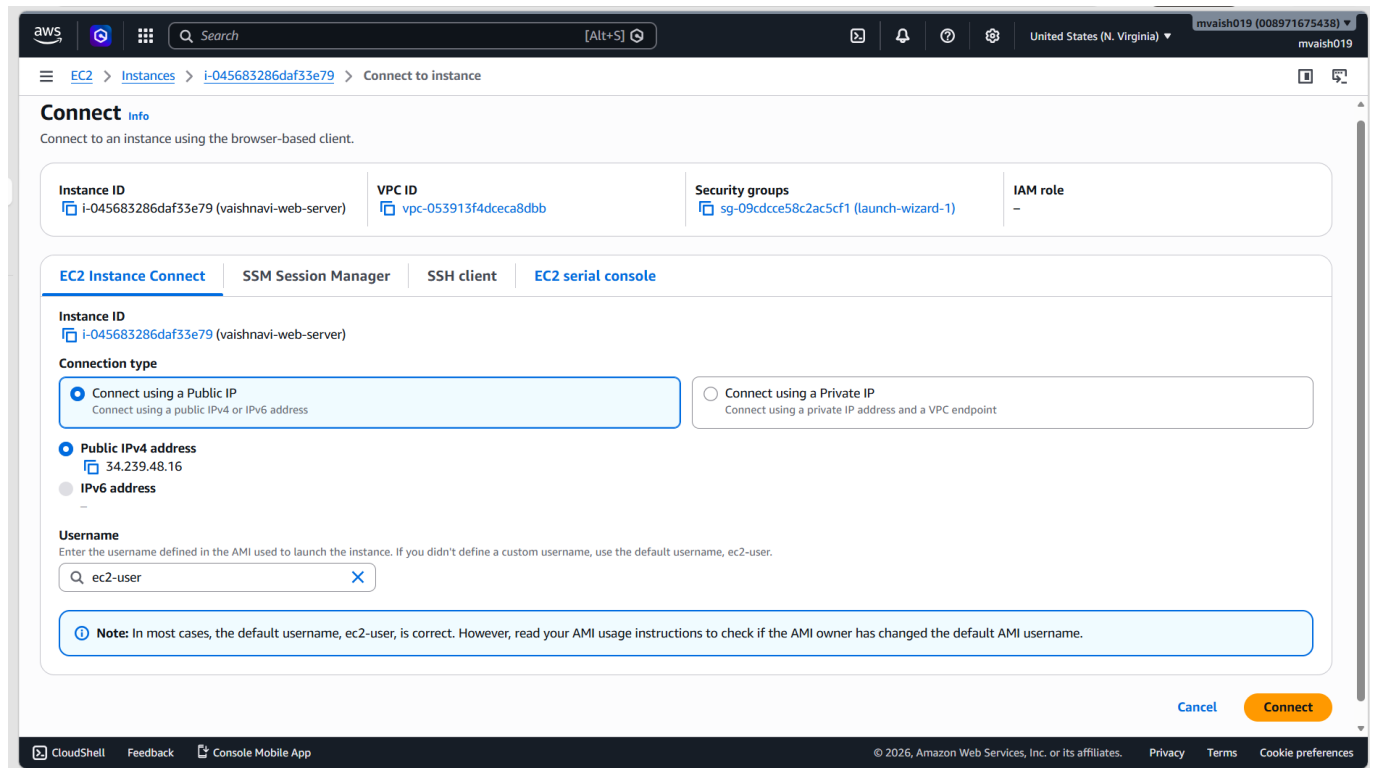
Step 1 – Launch EC2 Instance

An Amazon EC2 instance was launched to act as a virtual machine in the cloud. The instance was created using the Amazon Linux operating system and configured with a public IP address to allow internet access.



Step 2 – Connect to EC2 Instance

The EC2 Instance Connect feature was used to securely connect to the virtual machine through the AWS Management Console. This provided terminal access for installing and configuring the web server.



Step 3 – Install Apache Web Server

Commands Used

```
sudo dnf update -y
```

```
sudo dnf install httpd -y
```

```
sudo systemctl start httpd
```

```
sudo systemctl enable httpd
```

```
aws Search [Alt+S] mvaish019 (008971675438) mvaish019

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Wed Jun  3 07:13:24 2026 from 18.206.107.28
[ec2-user@ip-172-31-44-163 ~]$ history
 1 sudo yum install -y httpd
 2 sudo systemctl start httpd
 3 sudo systemctl enable httpd
 4 cd /var/www/html
 5 sudo nano index.html
 6 sudo nano /var/www/html/index.html
 7 history
[ec2-user@ip-172-31-44-163 ~]$ ^[[200~sudo systemctl status httpd~
-bash: $'\E[200~sudo': command not found
[ec2-user@ip-172-31-44-163 ~]$ history | grep httpd
 1 sudo yum install -y httpd
 2 sudo systemctl start httpd
 3 sudo systemctl enable httpd
 8 sudo systemctl status httpd~
 9 history | grep httpd
[ec2-user@ip-172-31-44-163 ~]$
```

i-045683286daf33e79 (vaishnavi-web-server)
PublicIPs: 34.239.48.16 PrivateIPs: 172.31.44.163

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Step 4 – Create Web Page

Command Used

```
sudo nano /var/www/html/index.html
```

A custom HTML webpage was created and stored in Apache's default web directory. This page acts as the home page of the hosted website.

```
aws Search [Alt+S] mvaish019 (008971675438) mvaish019

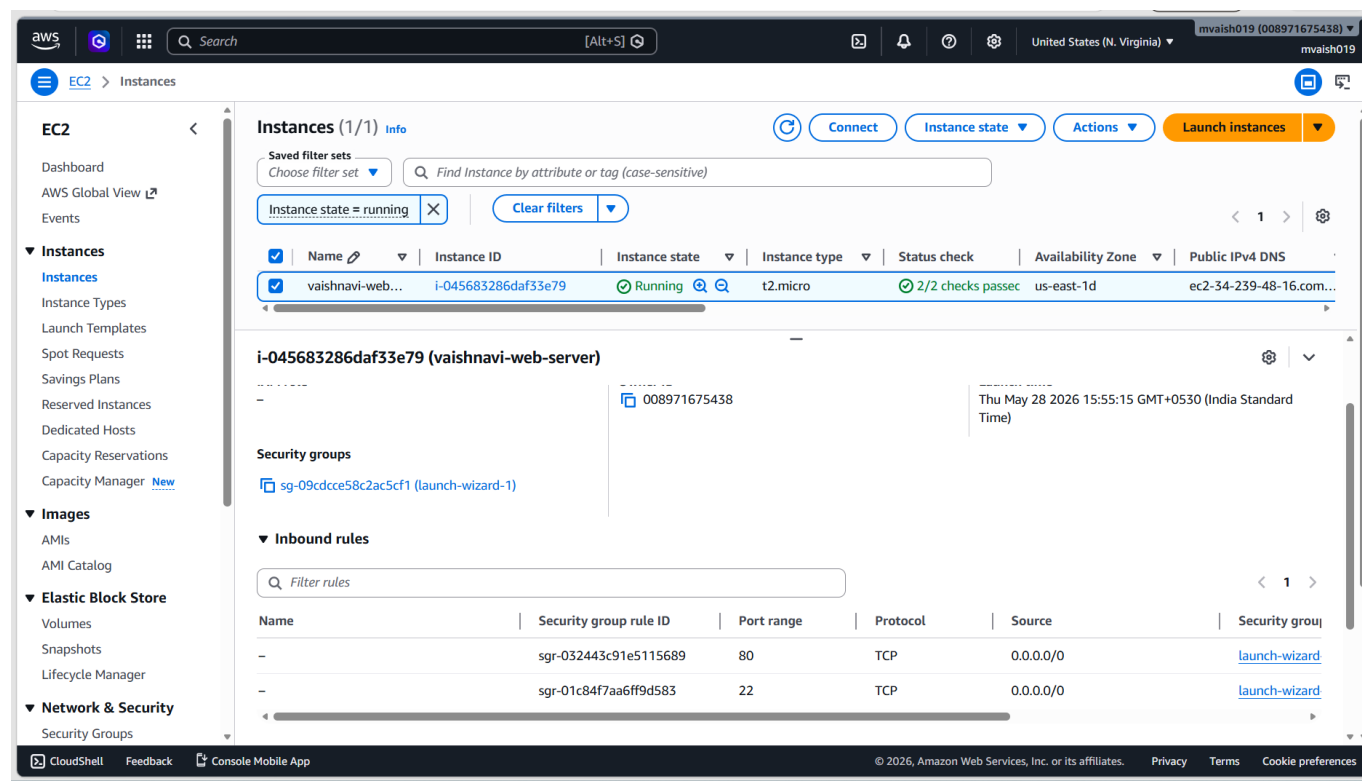
GNU nano 8.3 /var/www/html/index.html
<!DOCTYPE html>
<html>
<head>
  <title>Cloud VM Demo</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f4f6f9;
      text-align: center;
      padding-top: 100px;
    }
    .box {
      background: white;
      width: 60%;
      margin: auto;
      padding: 40px;
      border-radius: 10px;
      box-shadow: 0px 0px 15px rgba(0,0,0,0.1);
    }
    h1 {
      color: #0073e6;
    }
    p {
      color: #555;
      font-size: 18px;
    }
  </style>
</head>
<body>
<div class="box">
  <h1>Hello from Cloud VM 🚀</h1>
  <p>Web Server Successfully Hosted on AWS EC2</p>
  <p>Cloud Computing Internship - Alfido Tech</p>
</div>
</body>
</html>
```

i-045683286daf33e79 (vaishnavi-web-server)
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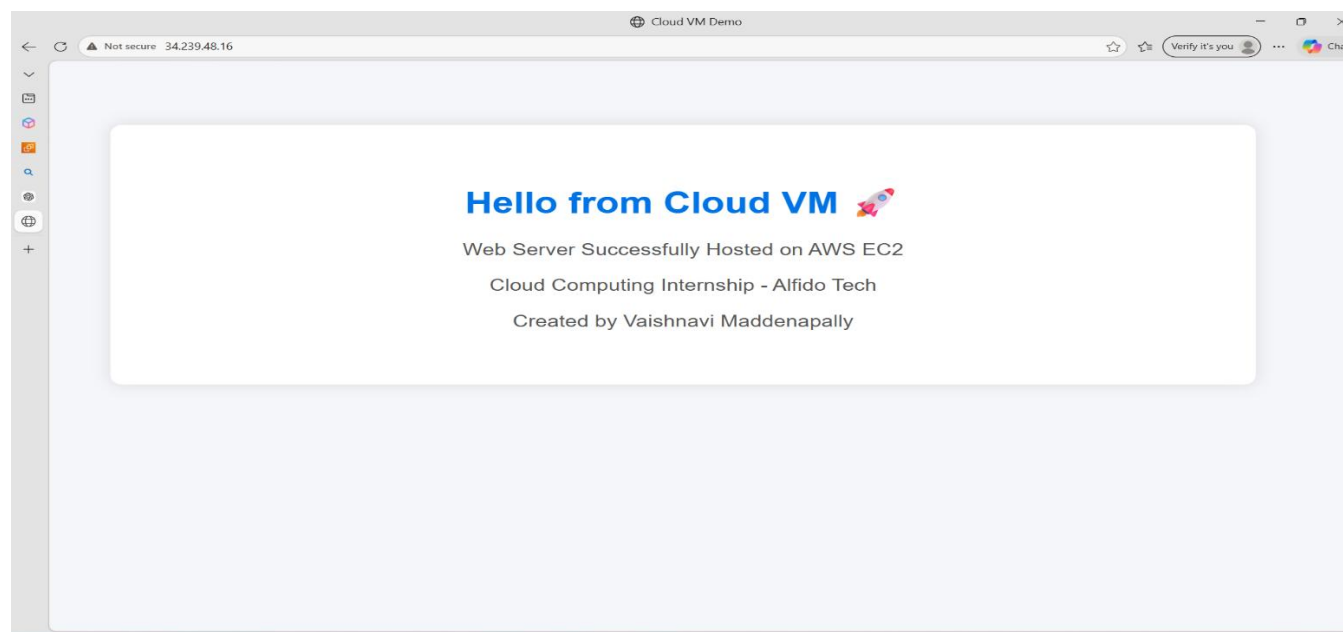
Step 5 – Configure Security Group

The EC2 Security Group was configured to allow HTTP traffic on Port 80. This enables users on the internet to access the hosted webpage.



Step 6 – Access Website Using Public IP

The public IP address of the EC2 instance was entered in a web browser. The hosted webpage was successfully displayed, confirming that the web server was working correctly.



Cloud VM Demo

Output

The web server was successfully deployed on an AWS EC2 virtual machine. The hosted webpage became publicly accessible using the instance's public IP address, demonstrating successful deployment and configuration of a cloud-based web server.

Conclusion

This task provided hands-on experience with cloud virtual machines, web server installation, Linux administration, and network security configuration. It demonstrated how cloud infrastructure can be used to host and serve web applications over the internet.